

IAL EVALUTION STINCT-2

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AI-Powered Internal Operations

Two enterprise automation projects purpose eliminating manual bottlenecks in crew operations at IndiGo Airlines.

The Manual Problem We Set Out to Solve

Two high-friction workflows were consuming critical hours of crew operations staff time — with error-prone, non-auditable outcomes.

Crew Relocation Routing

Manual flight booking, AIMS code application, and hotel logistics across a 3-day relocation window — tracked in spreadsheets with no automated validation.

URTI Leave Processing

URTI requests handled via email, requiring manual balance checks, roster verification, and leave rule interpretation — all without a consistent audit trail.

Crew Rostering Assignment Workflow

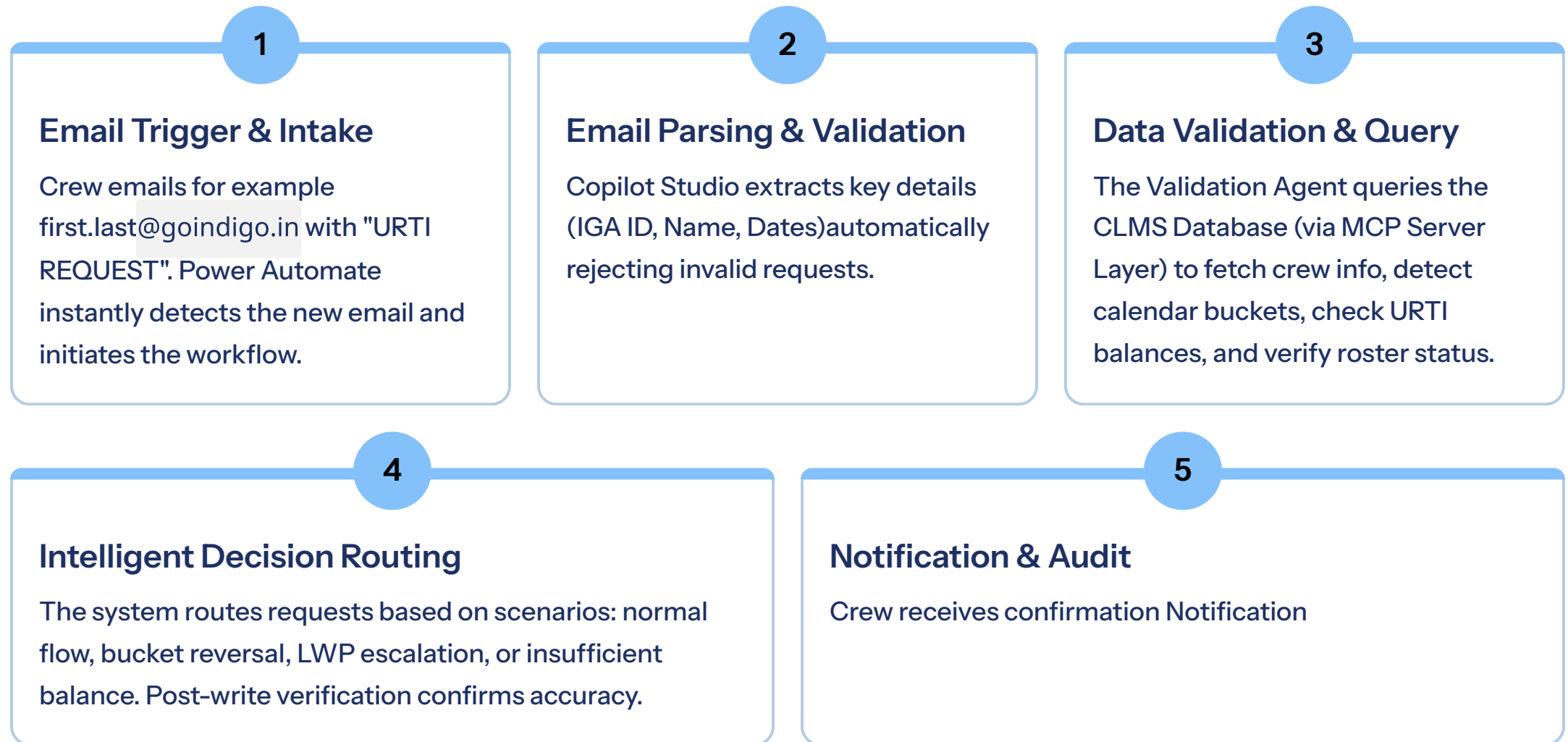
Our fully automated Python and FastAPI workflow transforms the manual processing of crew roster allocations into a streamlined, efficient, and auditable process. From email input to AIMS-ready output, this solution handles complex logic and edge cases, dramatically reducing processing time from hours to minutes.



This solution effectively manages complex scenarios like dummy flights for own-transport and provides backup flight selections, ensuring operational resilience.

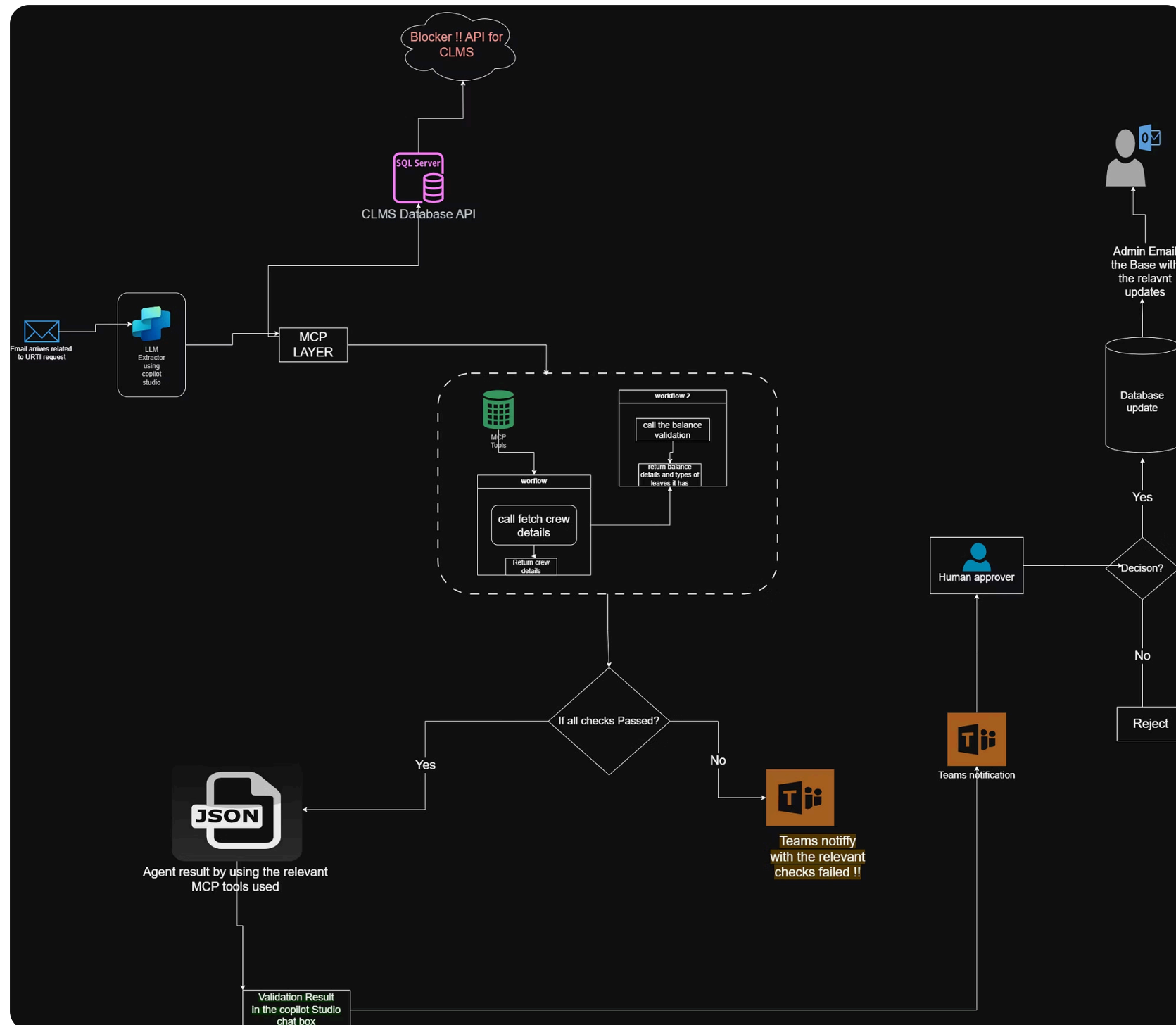
URTI: Email-Triggered Agentic Workflow

Our automated leave balance update process, powered by Copilot Studio agents, ensures efficient and accurate handling of URTI requests.



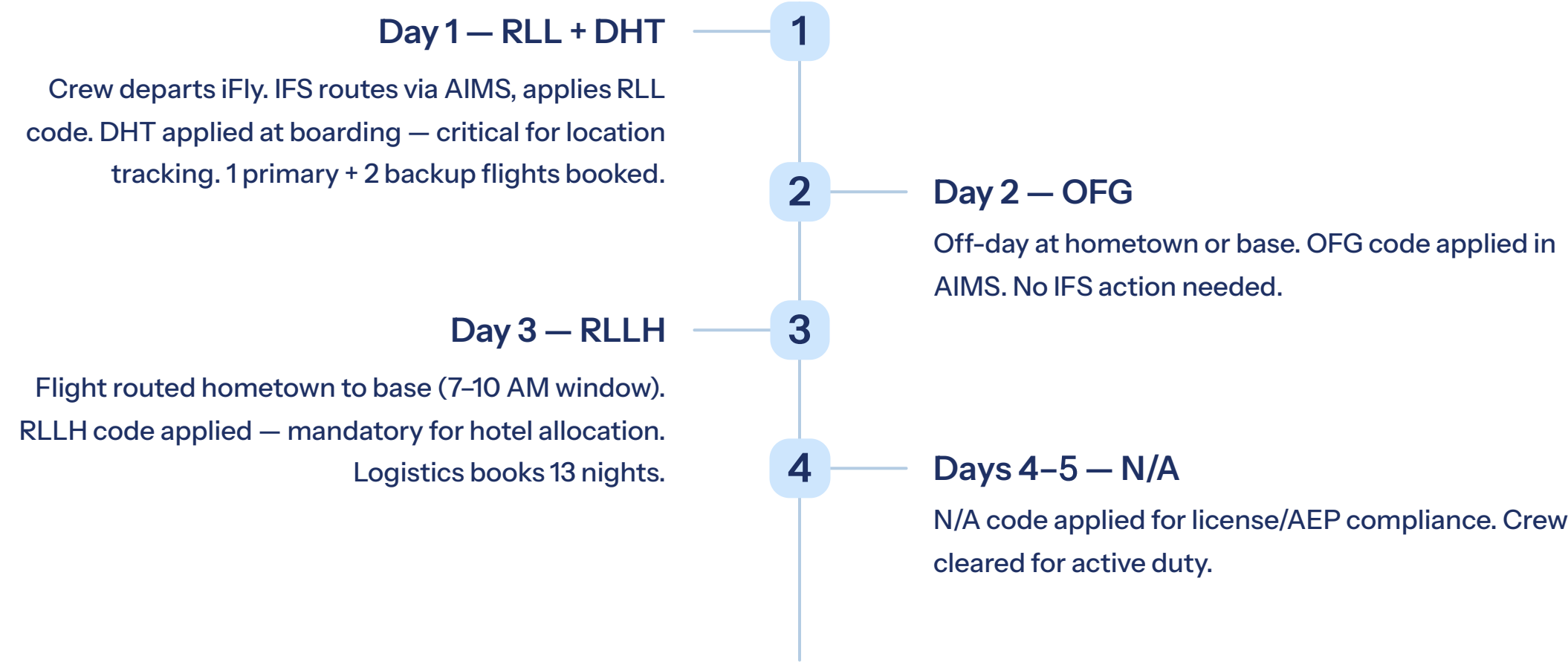
Architecture is Shared in next page

Architecture For Solution Diagram



Crew Relocation Pipeline: The 3-Day Process

From iFly final batch to active duty — every step automated, AIMS codes applied correctly, and edge cases handled without manual intervention.



Key Results & Impact

Both projects will deliver measurable operational improvements(if **not considering the blocker of the resources?**) and are aligned with IndiGo's compliance and standards.

01	02	03
Full Process	AIMS Codes	Automation
Full relocation routing end-to-end has been understand and implemented using the python and fastapi services.	RLL, RLLH, DHT, OFG, N/A — applied using the pipeline with all rules intact as it is.	Full URTI Process can be automate if the issue of the MCP, under the copilot studio agent is solved.

Key Learnings & Strategic Takeaways

→ **Agentic AI Works in Production**

Copilot Studio + Power Automate handles complex, rule-driven decisions reliably —with rollback built in from day one.

→ **Edge Cases Are the Real Product**

Own-transport crews, dummy flights, past-dated requests — the automation value is in handling what humans misses.

→ **AI workflows understanding**

Manual Workflow can be rewritten and been automate with the help of AI if done logically with taken care of all business requirements.

→ **Scalable Architecture for More Use Cases**

FastAPI microservices + Python pipelines provide a reusable pattern for future crew ops, IFS automation projects.

THANK YOU